

Progress Test 2.C — Elec-S1

2.C · Pouillet's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Tables often give resistivities in $\mu\Omega\cdot\text{cm}$. Convert $\rho = 1.7 \mu\Omega\cdot\text{cm}$ into $\Omega\cdot\text{m}$. Give the exact result.

$\rho = 1.7 \times 10^{-6} \Omega \times 10^{-2} \text{ m}$	$\rho = 1.7 \times 10^{-8} \Omega \text{ m}$	(the resistivity of copper)
---	--	-----------------------------

Marking

①
0 1

Marking scheme

① 1 pt $1.7 \times 10^{-8} \Omega \text{ m}$.

Exercise 2. Pouillet's law

(5 marks)

State Pouillet's law giving the resistance R of a conducting wire of length L and cross-sectional area S , made of a material of resistivity ρ . Give the SI unit of each of the four quantities. Define the conductivity σ and give its unit.

$R = \rho \frac{L}{S}$	a longer wire resists more, a thicker wire conducts more easily.
Units: R in ohms (Ω), ρ in $\Omega \text{ m}$, L in metres (m), S in square metres (m^2). (check: $\Omega \text{ m} \times \text{m}/\text{m}^2 = \Omega$)	
Conductivity: $\sigma = \frac{1}{\rho}$, in siemens per metre (S m^{-1}).	

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts $R = \rho L/S$ (0 if L and S are swapped).
- ② 2 pts The four units (1 pt lost per mistake).
- ③ 1 pt $\sigma = 1/\rho$ in S m^{-1} .

Examiner's tip

Checking the dimensions ($\Omega \text{ m} \times \text{m}/\text{m}^2 = \Omega$) catches a swapped L and S in two seconds.

Exercise 3. Resistivity and geometry

(3 marks)

A wire is replaced by a wire of the same material, twice as long and with twice the diameter. Its resistance is:

- ☐ multiplied by 2 ☐ unchanged ☒ divided by 2 ☐ divided by 4

Why: $L \rightarrow 2L$, and doubling the diameter multiplies $S = \pi d^2/4$ by 4, so $R \rightarrow R \times 2/4 = R/2$.

Among these materials, the best conductor is:

- ☐ iron ☐ aluminium ☐ copper ☒ silver

Why: it has the lowest resistivity: $\rho_{\text{Ag}} = 1.6 \times 10^{-8} \Omega \text{ m}$ (Cu: 1.7, Al: 2.8, Fe: 10×10^{-8}).

When a metallic wire heats up, its resistance:

- ☒ increases ☐ decreases ☐ stays the same ☐ vanishes

Why: for most metals the resistivity increases with temperature: a heated metal conducts slightly less well.

Exercise 4. Resistance of a copper wire

(5 marks)

A copper wire of length L and cross-sectional area S carries a current i . Determine its resistance R , then the power p it dissipates by the Joule effect.

$$L = 20 \text{ m} \quad S = 0.50 \text{ mm}^2 \quad \rho = 1.7 \times 10^{-8} \Omega \text{ m} \quad i = 5.0 \text{ A}$$

$$R = \rho \frac{L}{S} \quad \text{with } S = 0.50 \text{ mm}^2 = 0.50 \times (10^{-3} \text{ m})^2 = 5.0 \times 10^{-7} \text{ m}^2$$

$$R = \frac{1.7 \times 10^{-8} \Omega \text{ m} \times 20 \text{ m}}{5.0 \times 10^{-7} \text{ m}^2} = \frac{1.7 \times 20}{5.0} \times 10^{-8+7} \Omega = 6.8 \times 10^{-1} \Omega$$

$$R = 0.68 \Omega$$

$$p = Ri^2 = 0.68 \Omega \times (5.0 \text{ A})^2 = 0.68 \times 25 \text{ W} \quad p = 17 \text{ W}$$

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	
⑤		
0	1	

Marking scheme

- ① 1 pt Literal expression $R = \rho L/S$.
 ② 1 pt $S = 5.0 \times 10^{-7} \text{ m}^2$ ($1 \text{ mm}^2 = 10^{-6} \text{ m}^2$).
 ③ 1 pt $R = 0.68 \Omega$.
 ④ 1 pt Literal expression $p = Ri^2$.
 ⑤ 1 pt $p = 17 \text{ W}$, with its unit.

Examiner's tip

$\text{mm}^2 \rightarrow \text{m}^2$: square the conversion factor, $(10^{-3})^2 = 10^{-6}$. You got this right in Test 1; keep writing it out explicitly.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = / 20